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Inventor(s)	Hwang; Robin Christine et al.

High electron mobility transistor device and manufacturing method thereof

Abstract

A HEMT device including a substrate structure, a channel layer, a barrier layer, a gate electrode, a drain electrode, a first source field plate, a second source field plate, and a dielectric structure is provided. The first source field plate extends from the second side of the gate electrode to the first side of the gate electrode. The second source field plate is located on the first side of the gate electrode and is located between the drain electrode and the first source field plate. There is a gap between the first source field plate and the second source field plate. The first source field plate has an end adjacent to the gap. The thickness of the dielectric structure located directly below the second source field plate is greater than the thickness of the dielectric structure located directly below the end of the first source field plate.

Inventors: Hwang; Robin Christine (Taipei, TW), Chou; Jih-Wen (Hsinchu, TW), Chen; Hwi-Huang (Hsinchu, TW), Chen; Hsin-Hong (Hsinchu, TW), Huang; Yu-Jen (Hsinchu, TW), Lu; Chih-Hung (Taichung, TW)

Applicant: Powerchip Semiconductor Manufacturing Corporation (Hsinchu, TW)

Family ID: 89722693

Assignee: Powerchip Semiconductor Manufacturing Corporation (Hsinchu, TW)

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Primary Examiner: Tobergte; Nicholas J

Attorney, Agent or Firm: JCIPRNET

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims the priority benefit of Taiwan application serial no. 112104866, filed on Feb. 10, 2023. The entirety of the above-mentioned patent application is hereby incorporated by reference herein and made a part of this specification.

BACKGROUND

Technical Field

(2) The invention relates to a semiconductor device and a manufacturing method thereof, and particularly relates to a high electron mobility transistor (HEMT) device and a manufacturing

method thereof.

Description of Related Art

(3) The HEMT is a field effect transistor and can have higher breakdown voltage and higher reliability. However, how to further improve the electrical performance of the HEMT is the goal of continuous efforts.

SUMMARY

(4) The invention provides a HEMT device and a manufacturing method thereof, which can increase the breakdown voltage of the HEMT device.

(5) The invention provides a HEMT device, which includes a substrate structure, a channel layer, a barrier layer, a gate electrode, a drain electrode, a first source field plate, a second source field plate, and a dielectric structure. The channel layer is located on the substrate structure. The barrier layer is located on the channel layer. The gate electrode is located on the barrier layer. The gate electrode has a first side and a second side opposite to each other. The drain electrode is located on the first side of the gate electrode. The first source field plate extends from the second side of the gate electrode to the first side of the gate electrode. The second source field plate is located on the first side of the gate electrode and is located between the drain electrode and the first source field plate. There is a gap between the top-view pattern of the first source field plate and the top-view pattern of the second source field plate. The first source field plate has an end adjacent to the gap. The dielectric structure is located between the first source field plate and the gate electrode, between the first source field plate and the barrier layer, and between the second source field plate and the barrier layer. The thickness of the dielectric structure located directly below the second source field plate is greater than the thickness of the dielectric structure located directly below the end of the first source field plate.

(6) According to an embodiment of the invention, in the HEMT device, the first source field plate and the second source field plate may be derived from the same material layer.

(7) According to an embodiment of the invention, in the HEMT device, a portion of the first source field plate may be located directly above the gate electrode.

(8) According to an embodiment of the invention, in the HEMT device, the first source field plate and the second source field plate may be electrically connected to each other.

(9) According to an embodiment of the invention, in the HEMT device, the first source field plate may be directly connected to the second source field plate.

(10) According to an embodiment of the invention, in the HEMT device, the first source field plate and the second source field plate may be separated from each other.

(11) According to an embodiment of the invention, the HEMT device may further include an interconnect structure. The first source field plate and the second source field plate may be electrically connected to each other by the interconnect structure.

(12) According to an embodiment of the invention, in the HEMT device, the dielectric structure may include a first dielectric layer, a second dielectric layer, and an etch stop layer. The first dielectric layer is located on the barrier layer. The second dielectric layer is located on the first dielectric layer. The etch stop layer is located between the first dielectric layer and the second dielectric layer.

(13) According to an embodiment of the invention, in the HEMT device, the first dielectric layer and the etch stop layer may be located directly below the first source field plate. The second dielectric layer is not located directly below the first source field plate. The first dielectric layer, the etch stop layer, and the second dielectric layer may be located directly below the second source field plate.

(14) According to an embodiment of the invention, in the HEMT device, the drain electrode may be connected to the channel layer.

(15) According to an embodiment of the invention, in the HEMT device, the first source field plate located on the second side of the gate electrode may be connected to the channel layer.

(16) According to an embodiment of the invention, the HEMT device may further include a P-type gallium nitride (P-type GaN (pGaN)) layer. The P-type gallium nitride layer is located between the gate electrode and the barrier layer.

(17) According to an embodiment of the invention, in the HEMT device, the substrate structure may include a substrate, a nucleation layer, and a buffer layer. The nucleation layer is located on the substrate. The buffer layer is located on the nucleation layer.

(18) The invention provides a manufacturing method of a HEMT device, which includes the following steps. A substrate structure is provided. A channel layer is formed on the substrate structure. A barrier layer is formed on the channel layer. A gate electrode is formed on the barrier layer. The gate electrode has a first side and a second side opposite to each other. A drain electrode, a first source field plate, and a second source field plate are formed. The drain electrode is located on the first side of the gate electrode. The first source field plate extends from the second side of the gate electrode to the first side of the gate electrode. The second source field plate is located on the first side of the gate electrode and is located between the drain electrode and the first source field plate. There is a gap between the top-view pattern of the first source field plate and the top-view pattern of the second source field plate. The first source field plate has an end adjacent to the gap. A dielectric structure is formed between the first source field plate and the gate electrode, between the first source field plate and the barrier layer, and between the second source field plate and the barrier layer. The thickness of the dielectric structure located directly below the second source field plate is greater than the thickness of the dielectric structure located directly below the end of the first source field plate.

(19) According to an embodiment of the invention, in the manufacturing method of the HEMT device, the first source field plate and the second source field plate may be simultaneously formed by the same metal process.

(20) According to an embodiment of the invention, in the manufacturing method of the HEMT device, the dielectric structure may include a first dielectric layer, a second dielectric layer, and an etch stop layer. The first dielectric layer is located on the barrier layer. The second dielectric layer is located on the first dielectric layer. The etch stop layer is located between the first dielectric layer and the second dielectric layer.

(21) According to an embodiment of the invention, in the manufacturing method of the HEMT device, the first dielectric layer and the etch stop layer may be located directly below the first source field plate. The second dielectric layer is not located directly below the first source field plate. The first dielectric layer, the etch stop layer, and the second dielectric layer may be located directly below the second source field plate.

(22) According to an embodiment of the invention, in the manufacturing method of the HEMT device, the method of forming the barrier layer, the first dielectric layer, the etch stop layer, and the second dielectric layer may include the following steps. A barrier material layer is formed on the channel layer. The gate electrode is formed on the barrier material layer. A first dielectric material layer is formed on the barrier material layer and the gate electrode. An etch stop material layer is formed on the first dielectric material layer. A second dielectric material layer is formed on the etch stop material layer. The second dielectric material layer is patterned to form the second dielectric layer and to expose the etch stop material layer. The etch stop material layer, the first dielectric material layer, and the barrier material layer are patterned to form the etch stop layer, the first dielectric layer, and the barrier layer and to expose the channel layer.

(23) According to an embodiment of the invention, in the manufacturing method of the HEMT device, the method of forming the drain electrode, the first source field plate, and the second source field plate may include the following steps. A conductive material layer is formed on the channel layer and the dielectric structure. The conductive material layer is patterned to form the drain electrode, the first source field plate, and the second source field plate.

(24) According to an embodiment of the invention, the manufacturing method of the HEMT device

may further include the following step. A P-type gallium nitride layer is formed between the gate electrode and the barrier layer.

(25) Based on the above description, in the HEMT device and the manufacturing method thereof according to the invention, the second source field plate is located between the drain electrode and the first source field plate. There is a gap between the top-view pattern of the first source field plate and the top-view pattern of the second source field plate. The first source field plate has an end adjacent to the gap. The thickness of the dielectric structure located directly below the second source field plate is greater than the thickness of the dielectric structure located directly below the end of the first source field plate. Therefore, the breakdown voltage of the HEMT device according to the invention can be increased. In addition, when the reliability test is performed on the HEMT device according to the invention, the dispersion of the turn-on resistance ($R_{sub.on}$) of the HEMT device under multiple switching operations can be more concentrated. In this way, the HEMT device according to the invention can have better dynamic performance.

(26) In order to make the aforementioned and other objects, features and advantages of the invention comprehensible, several exemplary embodiments accompanied with drawings are described in detail below.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The accompanying drawings are included to provide a further understanding of the invention, and are incorporated in and constitute a part of this specification. The drawings illustrate embodiments of the invention and, together with the description, serve to explain the principles of the invention.

(2) FIG. 1A to FIG. 1H are cross-sectional views of a manufacturing process of a HEMT device according to some embodiments of the invention.

(3) FIG. 2 is a top view of the HEMT device in FIG. 1H.

(4) FIG. 3 is a top view of a HEMT device according to other embodiments of the invention.

DESCRIPTION OF THE EMBODIMENTS

(5) The embodiments are described in detail below with reference to the accompanying drawings, but the embodiments are not intended to limit the scope of the invention. For the sake of easy understanding, the same components in the following description will be denoted by the same reference symbols. In addition, the drawings are for illustrative purposes only and are not drawn to the original dimensions. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

(6) FIG. 1A to FIG. 1H are cross-sectional views of a manufacturing process of a HEMT device according to some embodiments of the invention. FIG. 2 is a top view of the HEMT device in FIG. 1H. FIG. 1A to FIG. 1H are cross-sectional views taken along section line I-I' in FIG. 2. In the top view of the present embodiment, some components in the cross-sectional view are omitted to clearly illustrate the configuration relationship between the components in the top view.

(7) Referring to FIG. 1A, a substrate structure **100** is provided. In some embodiments, the substrate structure **100** may include a substrate **102**, a nucleation layer **104**, and a buffer layer **106**. In some embodiments, the substrate **102** may be a semiconductor substrate such as a silicon substrate. The nucleation layer **104** is located on the substrate **102**. In some embodiments, the material of the nucleation layer **104** is, for example, aluminum nitride (AlN). In some embodiments, the method of forming the nucleation layer **104** is, for example, an epitaxial growth method. The buffer layer **106** is located on the nucleation layer **104**. In some embodiments, the material of the buffer layer **106** is, for example, aluminum gallium nitride (AlGaN) or aluminum nitride. In some embodiments, the method of forming the buffer layer **106** is, for example, an epitaxial growth method.

(8) A channel layer **108** is formed on the substrate structure **100**. In some embodiments, the channel layer **108** may be formed on the buffer layer **106**. In some embodiments, the material of the channel layer **108** is, for example, gallium nitride (GaN). In some embodiments, the method of forming the channel layer **108** is, for example, an epitaxial growth method.

(9) A barrier material layer **110** may be formed on the channel layer **108**. In some embodiments, the material of the barrier material layer **110** is, for example, aluminum gallium nitride. In some embodiments, the method of forming the barrier material layer **110** is, for example, an epitaxial growth method.

(10) A gate electrode **112** may be formed on the barrier material layer **110**. The gate electrode **112** has a first side S1 and a second side S2 opposite to each other. In some embodiments, the material of the gate electrode **112** is, for example, titanium nitride (TiN), but the invention is not limited thereto. In some embodiments, a P-type gallium nitride layer **114** may be formed between the gate electrode **112** and the barrier material layer **110**.

(11) In some embodiments, the method of forming the gate electrode **112** and the P-type gallium nitride layer **114** may include the following steps. First, a P-type gallium nitride material layer (not shown) and a gate electrode material layer (not shown) may be sequentially formed on the barrier material layer **110**. In some embodiments, the method of forming the P-type gallium nitride material layer is, for example, an epitaxial growth method. In some embodiments, the method of forming the gate electrode material layer is, for example, a chemical vapor deposition (CVD) method or a physical vapor deposition (PVD) method. Then, the gate electrode material layer and the P-type gallium nitride material layer may be patterned by a lithography process and an etching process (e.g., dry etching process) to form the gate electrode **112** and the P-type gallium nitride layer **114**.

(12) A dielectric material layer **116** may be formed on the barrier material layer **110** and the gate electrode **112**. The dielectric material layer **116** may cover the barrier material layer **110**, the gate electrode **112**, and the P-type gallium nitride layer **114**. In some embodiments, the material of the dielectric material layer **116** is, for example, silicon oxide or silicon nitride. In some embodiments, the method of forming the dielectric material layer **116** is, for example, a CVD method.

(13) An etch stop material layer **118** may be formed on the dielectric material layer **116**. In some embodiments, the material of the etch stop material layer **118** is, for example, aluminum oxide (Al.sub.2O.sub.3) or aluminum nitride. In some embodiments, the method of forming the etch stop material layer **118** is, for example, an atomic layer deposition (ALD) method.

(14) Referring to FIG. 1B, a dielectric material layer **120** may be formed on the etch stop material layer **118**. In some embodiments, the material of the dielectric material layer **120** is, for example, silicon oxide or silicon nitride. In some embodiments, the method of forming the dielectric material layer **120** is, for example, a CVD method.

(15) Referring to FIG. 1C, a patterned photoresist layer **122** may be formed on the dielectric material layer **120**. In some embodiments, the patterned photoresist layer **122** may be formed by a lithography process.

(16) Referring to FIG. 1D, a portion of the dielectric material layer **120** may be removed by using the patterned photoresist layer **122** as a mask to form a dielectric layer **120a**. For example, a dry etching process may be performed on the dielectric material layer **120** by using the patterned photoresist layer **122** as a mask and using the etch stop material layer **118** as an etch stop layer to remove the portion of the dielectric material layer **120**. Therefore, the dielectric material layer **120** may be patterned to form the dielectric layer **120a** and to expose the etch stop material layer **118**. In some embodiments, the material of the dielectric layer **120a** is, for example, silicon oxide or silicon nitride.

(17) Referring to FIG. 1E, the patterned photoresist layer **122** may be removed. In some embodiments, the method of removing the patterned photoresist layer **122** is, for example, a dry stripping method or a wet stripping method.

(18) A patterned photoresist layer **124** may be formed on the etch stop material layer **118** and the dielectric layer **120a**. In some embodiments, the patterned photoresist layer **124** may be formed by a lithography process.

(19) Referring to FIG. 1F, a portion of the etch stop material layer **118**, a portion of the dielectric material layer **116**, and a portion of the barrier material layer **110** may be removed by using the patterned photoresist layer **124** as a mask to form an etch stop layer **118a**, a dielectric layer **116a**, and a barrier layer **110a**. For example, a dry etching process may be performed on the etch stop material layer **118**, the dielectric material layer **116**, and the barrier material layer **110** by using the patterned photoresist layer **124** as a mask and using the channel layer **108** as an etch stop layer to remove the portion of the etch stop material layer **118**, the portion of the dielectric material layer **116**, and the portion of the barrier material layer **110**. Therefore, the etch stop material layer **118**, the dielectric material layer **116**, and the barrier material layer **110** may be patterned to form the etch stop layer **118a**, the dielectric layer **116a**, and the barrier layer **110a** and to expose the channel layer **108**. In some embodiments, the material of the etch stop layer **118a** is, for example, aluminum oxide or aluminum nitride. In some embodiments, the material of the dielectric layer **116a** is, for example, silicon oxide or silicon nitride. In some embodiments, the material of the barrier layer **110a** is, for example, aluminum gallium nitride.

(20) By the above method, the barrier layer **110a** may be formed on the channel layer **108**, the gate electrode **112** may be formed on the barrier layer **110a**, the P-type gallium nitride layer **114** may be formed between the gate electrode **112** and the barrier layer **110a**, and the dielectric structure **126** may be formed. In some embodiments, the dielectric structure **126** may include the dielectric layer **116a**, the dielectric layer **120a**, and the etch stop layer **118a**. The dielectric layer **116a** is located on the barrier layer **110a**. The dielectric layer **116a** may cover the gate electrode **112** and the P-type gallium nitride layer **114**. The dielectric layer **120a** is located on the dielectric layer **116a**. The etch stop layer **118a** is located between the dielectric layer **116a** and the dielectric layer **120a**.

(21) Referring to FIG. 1G, the patterned photoresist layer **124** may be removed. In some embodiments, the method of removing the patterned photoresist layer **124** is, for example, a dry stripping method or a wet stripping method.

(22) A conductive material layer **128** may be formed on the channel layer **108** and the dielectric structure **126**. In some embodiments, the material of the conductive material layer **128** is, for example, a conductive material that may form an ohmic contact with the channel layer **108**. In some embodiments, the material of the conductive material layer **128** is, for example, titanium, aluminum copper alloy (AlCu), or a combination thereof. In some embodiments, the method of forming the conductive material layer **128** is, for example, a CVD method or a PVD method.

(23) Referring to FIG. 1H and FIG. 2, the conductive material layer **128** may be patterned to form a drain electrode **128a**, a source field plate **128b**, and a source field plate **128c**. The drain electrode **128a** is located on the first side S1 of the gate electrode **112**. The source field plate **128b** extends from the second side S2 of the gate electrode **112** to the first side S1 of the gate electrode **112**. The source field plate **128b** may pass over the gate electrode **112**. The source field plate **128c** is located on the first side S1 of the gate electrode **112** and is located between the drain electrode **128a** and the source field plate **128b**. There is a gap G1 between the top-view pattern of the source field plate **128b** and the top-view pattern of the source field plate **128c**. The source field plate **128b** has an end E1 adjacent to the gap G1. In some embodiments, the source field plate **128b** and the source field plate **128c** may be derived from the same material layer (e.g., conductive material layer **128**). In some embodiments, the source field plate **128b** and the source field plate **128c** may be simultaneously formed by the same metal process. In some embodiments, the drain electrode **128a**, the source field plate **128b**, and the source field plate **128c** may be derived from the same material layer (e.g., conductive material layer **128**). In some embodiments, the drain electrode **128a**, the source field plate **128b**, and the source field plate **128c** may be simultaneously formed by the same metal process. For example, after the conductive material layer **128** is formed, the conductive

material layer **128** may be patterned by a lithography process and an etching process (e.g., dry etching process) to simultaneously form the drain electrode **128a**, the source field plate **128b**, and the source field plate **128c**. In some embodiments, the materials of the drain electrode **128a** and the source field plate **128b** is, for example, a conductive material that may form an ohmic contact with the channel layer **108**. In some embodiments, the materials of the drain electrode **128a**, the source field plate **128b**, and the source field plate **128c** are, for example, titanium, aluminum copper alloy, or a combination thereof.

(24) In addition, by the above method, the dielectric structure **126** may be formed between the source field plate **128b** and the gate electrode **112**, between the source field plate **128b** and the barrier layer **110a**, and between the source field plate **128c** and the barrier layer **110a**. The thickness **T1** of the dielectric structure **126** located directly below the source field plate **128c** is greater than the thickness **T2** of the dielectric structure **126** located directly below the end **E1** of the source field plate **128b**. In some embodiments, the dielectric layer **116a** and the etch stop layer **118a** may be located directly below the source field plate **128b**. The dielectric layer **120a** is not located directly below the source field plate **128b**. The dielectric layer **116a**, the etch stop layer **118a**, and the dielectric layer **120a** may be located directly below the source field plate **128c**. Therefore, the thickness **T1** of the dielectric structure **126** located directly below the source field plate **128c** may be greater than the thickness **T2** of the dielectric structure **126** located directly below the end **E1** of the source field plate **128b**.

(25) Hereinafter, the HEMT device **10** of the above embodiment is described with reference to FIG. **1H** and FIG. **2**. In addition, although the method for forming the HEMT device **10** is described by taking the above method as an example, the invention is not limited thereto.

(26) Referring to FIG. **1H** and FIG. **2**, the HEMT device **10** includes a substrate structure **100**, a channel layer **108**, a barrier layer **110a**, a gate electrode **112**, a drain electrode **128a**, a source field plate **128b**, a source field plate **128c**, and a dielectric structure **126**. In some embodiments, the substrate structure **100** may include a substrate **102**, a nucleation layer **104**, and a buffer layer **106**. The nucleation layer **104** is located on the substrate **102**. The buffer layer **106** is located on the nucleation layer **104**. The channel layer **108** is located on the substrate structure **100**. The barrier layer **110a** is located on the channel layer **108**. The gate electrode **112** is located on the barrier layer **110a**. The gate electrode **112** has a first side **S1** and a second side **S2** opposite to each other. In some embodiments, the HEMT device **10** may further include a P-type gallium nitride layer **114**. The P-type gallium nitride layer **114** is located between the gate electrode **112** and the barrier layer **110a**.

(27) The drain electrode **128a** is located on the first side **S1** of the gate electrode **112**. In some embodiments, a portion of the drain electrode **128a** may extend onto the top surface of the dielectric structure **126**. In some embodiments, the drain electrode **128a** may be connected to the channel layer **108**.

(28) The source field plate **128b** extends from the second side **S2** of the gate electrode **112** to the first side **S1** of the gate electrode **112**. The source field plate **128b** may pass over the gate electrode **112**. The portion **P1** of the source field plate **128b** located on the second side **S2** of the gate electrode **112** may be used as a source electrode. In some embodiments, the source field plate **128b** located on the second side **S2** of the gate electrode **112** may be connected to the channel layer **108**. In some embodiments, a portion of the source field plate **128b** may be located directly above the gate electrode **112**.

(29) The source field plate **128c** is located on the first side **S1** of the gate electrode **112** and is located between the drain electrode **128a** and the source field plate **128b**. There is a gap **G1** between the top-view pattern of the source field plate **128b** and the top-view pattern of the source field plate **128c**. The source field plate **128b** has an end **E1** adjacent to the gap **G1**. In some embodiments, the vertical projection of the entire source field plate **128c** does not fall on the source field plate **128b**. That is, the top-view pattern of the entire source field plate **128c** may not overlap

the top-view pattern of the source field plate **128b**. In some embodiments, the source field plate **128b** and the source field plate **128c** may be derived from the same material layer (e.g., conductive material layer **128** of FIG. 1G). In some embodiments, the drain electrode **128a**, the source field plate **128b**, and the source field plate **128c** may be derived from the same material layer (e.g., conductive material layer **128** of FIG. 1G).

(30) The source field plate **128b** and the source field plate **128c** may be electrically connected to each other. In some embodiments, as shown in FIG. 2, the source field plate **128b** may be directly connected to the source field plate **128c**. In some embodiments, as shown in FIG. 2, the source field plate **128b** and the source field plate **128c** may be integrally formed.

(31) The dielectric structure **126** is located between the source field plate **128b** and the gate electrode **112**, between the source field plate **128b** and the barrier layer **110a**, and between the source field plate **128c** and the barrier layer **110a**. In addition, the dielectric structure **126** may be further located between the source field plate **128b** and the P-type gallium nitride layer **114**. The thickness **T1** of the dielectric structure **126** located directly below the source field plate **128c** is greater than the thickness **T2** of the dielectric structure **126** located directly below the end **E1** of the source field plate **128b**. Therefore, the breakdown voltage of the HEMT device **10** can be increased. In addition, when the reliability test is performed on the HEMT device **10**, the dispersion of the turn-on resistance ($R_{sub.on}$) of the HEMT device **10** under multiple switching operations can be more concentrated. In this way, the HEMT device **10** can have better dynamic performance.

(32) In some embodiments, the dielectric structure **126** may include a dielectric layer **116a**, a dielectric layer **120a**, and an etch stop layer **118a**. The dielectric layer **116a** is located on the barrier layer **110a**. The dielectric layer **116a** may cover the gate electrode **112** and the P-type gallium nitride layer **114**. The dielectric layer **120a** is located on the dielectric layer **116a**. The etch stop layer **118a** is located between the dielectric layer **116a** and the dielectric layer **120a**. In some embodiments, the dielectric layer **116a** and the etch stop layer **118a** may be located directly below the source field plate **128b**. The dielectric layer **120a** is not located directly below the source field plate **128b**. The dielectric layer **116a**, the etch stop layer **118a**, and the dielectric layer **120a** may be located directly below the source field plate **128c**. Therefore, the thickness **T1** of the dielectric structure **126** located directly below the source field plate **128c** may be greater than the thickness **T2** of the dielectric structure **126** located directly below the end **E1** of the source field plate **128b**.

(33) In addition, the details (e.g., the material and the forming method) of each component in the HEMT device **10** have been described in detail in the above embodiments, and the description thereof is not repeated here.

(34) Based on the above embodiments, in the HEMT device **10** and the manufacturing method thereof, the source field plate **128c** is located between the drain electrode **128a** and the source field plate **128b**. There is a gap **G1** between the top-view pattern of the source field plate **128b** and the top-view pattern of the source field plate **128c**. The source field plate **128b** has an end **E1** adjacent to the gap **G1**. The thickness **T1** of the dielectric structure **126** located directly below the source field plate **128c** is greater than the thickness **T2** of the dielectric structure **126** located directly below the end **E1** of the source field plate **128b**. Therefore, the breakdown voltage of the HEMT device **10** can be increased. In addition, when the reliability test is performed on the HEMT device **10**, the dispersion of the turn-on resistance ($R_{sub.on}$) of the HEMT device **10** under multiple switching operations can be more concentrated. In this way, the HEMT device **10** can have better dynamic performance.

(35) FIG. 3 is a top view of a HEMT device according to other embodiments of the invention.

(36) Referring to FIG. 2 and FIG. 3, the difference between the HEMT device **20** of FIG. 3 and the HEMT device **10** of FIG. 2 is as follows. In the HEMT device **20** of FIG. 3, the source field plate **128b** and the source field plate **128c** may be separated from each other. In addition, the HEMT device **20** may further include an interconnect structure **130**. The source field plate **128b** and the source field plate **128c** may be electrically connected to each other by the interconnect structure

130. In FIG. 3, the interconnect structure **130** is represented by a line to simplify the figure. In some embodiments, the interconnect structure **130** may include a contact, a conductive line, or a combination thereof.

(37) In some embodiments, the HEMT device **20** may have a cross-sectional structure as shown in FIG. 1H. Therefore, in the HEMT device **20**, the thickness **T1** of the dielectric structure **126** located directly below the source field plate **128c** is greater than the thickness **T2** of the dielectric structure **126** located directly below the end **E1** of the source field plate **128b**.

(38) In addition, in the HEMT device **10** of FIG. 2 and the HEMT device **20** of FIG. 3, the same or similar components are denoted by the same reference symbols, and the description thereof is omitted. Furthermore, the details (e.g., the material and the forming method) of each component in the HEMT device **20** may refer to the description of the embodiment of the HEMT device **10**, and the description thereof is omitted here.

(39) Based on the above embodiments, in the HEMT device **20** and the manufacturing method thereof, the source field plate **128c** is located between the drain electrode **128a** and the source field plate **128b**. There is a gap **G1** between the top-view pattern of the source field plate **128b** and the top-view pattern of the source field plate **128c**. The source field plate **128b** has an end **E1** adjacent to the gap **G1**. The thickness **T1** of the dielectric structure **126** located directly below the source field plate **128c** is greater than the thickness **T2** of the dielectric structure **126** located directly below the end **E1** of the source field plate **128b**. Therefore, the breakdown voltage of the HEMT device **20** can be increased. In addition, when the reliability test is performed on the HEMT device **20**, the dispersion of the turn-on resistance ($R_{sub.on}$) of the HEMT device **20** under multiple switching operations can be more concentrated. In this way, the HEMT device **20** can have better dynamic performance.

(40) In summary, the HEMT device and the manufacturing method thereof of the aforementioned embodiments can increase the breakdown voltage of the HEMT device. In addition, when the reliability test is performed on the HEMT device of the aforementioned embodiments, the dispersion of the turn-on resistance ($R_{sub.on}$) of the HEMT device under multiple switching operations can be more concentrated. In this way, the HEMT device of the aforementioned embodiments can have better dynamic performance.

(41) Although the invention has been described with reference to the above embodiments, it will be apparent to one of ordinary skill in the art that modifications to the described embodiments may be made without departing from the spirit of the invention. Accordingly, the scope of the invention is defined by the attached claims not by the above detailed descriptions.

Claims

1. A high electron mobility transistor (HEMT) device, comprising: a substrate structure; a channel layer located on the substrate structure; a barrier layer located on the channel layer; a gate electrode located on the barrier layer and having a first side and a second side opposite to each other; a drain electrode located on the first side of the gate electrode; a first source field plate extending from the second side of the gate electrode to the first side of the gate electrode; a second source field plate located on the first side of the gate electrode and located between the drain electrode and the first source field plate, wherein there is a gap between a top-view pattern of the first source field plate and a top-view pattern of the second source field plate, and the first source field plate has an end adjacent to the gap; and a dielectric structure located between the first source field plate and the gate electrode, between the first source field plate and the barrier layer, and between the second source field plate and the barrier layer, wherein a thickness of the dielectric structure located directly below the second source field plate is greater than a thickness of the dielectric structure located directly below the end of the first source field plate.

2. The HEMT device according to claim 1, wherein the first source field plate and the second

source field plate are derived from the same material layer.

3. The HEMT device according to claim 1, wherein a portion of the first source field plate is located directly above the gate electrode.

4. The HEMT device according to claim 1, wherein the first source field plate and the second source field plate are electrically connected to each other.

5. The HEMT device according to claim 1, wherein the first source field plate is directly connected to the second source field plate.

6. The HEMT device according to claim 1, wherein the first source field plate and the second source field plate are separated from each other.

7. The HEMT device according to claim 6, further comprising: an interconnect structure, wherein the first source field plate and the second source field plate are electrically connected to each other by the interconnect structure.

8. The HEMT device according to claim 1, wherein the dielectric structure comprises: a first dielectric layer located on the barrier layer; a second dielectric layer located on the first dielectric layer; and an etch stop layer located between the first dielectric layer and the second dielectric layer.

9. The HEMT device according to claim 8, wherein the first dielectric layer and the etch stop layer are located directly below the first source field plate, the second dielectric layer is not located directly below the first source field plate, and the first dielectric layer, the etch stop layer, and the second dielectric layer are located directly below the second source field plate.

10. The HEMT device according to claim 1, wherein the drain electrode is connected to the channel layer.

11. The HEMT device according to claim 1, wherein the first source field plate located on the second side of the gate electrode is connected to the channel layer.

12. The HEMT device according to claim 1, further comprising: a P-type gallium nitride layer located between the gate electrode and the barrier layer.

13. The HEMT device according to claim 1, wherein the substrate structure comprises: a substrate; a nucleation layer located on the substrate; and a buffer layer located on the nucleation layer.

14. A manufacturing method of a HEMT device, comprising: providing a substrate structure; forming a channel layer on the substrate structure; forming a barrier layer on the channel layer; forming a gate electrode on the barrier layer, wherein the gate electrode has a first side and a second side opposite to each other; forming a drain electrode, a first source field plate, and a second source field plate, wherein the drain electrode is located on the first side of the gate electrode, and the first source field plate extends from the second side of the gate electrode to the first side of the gate electrode, the second source field plate is located on the first side of the gate electrode and is located between the drain electrode and the first source field plate, there is a gap between a top-view pattern of the first source field plate and a top-view pattern of the second source field plate, and the first source field plate has an end adjacent to the gap; and forming a dielectric structure between the first source field plate and the gate electrode, between the first source field plate and the barrier layer, and between the second source field plate and the barrier layer, wherein a thickness of the dielectric structure located directly below the second source field plate is greater than a thickness of the dielectric structure located directly below the end of the first source field plate.

15. The manufacturing method of the HEMT device according to claim 14, wherein the first source field plate and the second source field plate are simultaneously formed by the same metal process.

16. The manufacturing method of the HEMT device according to claim 14, wherein the dielectric structure comprises: a first dielectric layer located on the barrier layer; a second dielectric layer located on the first dielectric layer; and an etch stop layer located between the first dielectric layer and the second dielectric layer.

17. The manufacturing method of the HEMT device according to claim 16, wherein the first

dielectric layer and the etch stop layer are located directly below the first source field plate, the second dielectric layer is not located directly below the first source field plate, and the first dielectric layer, the etch stop layer, and the second dielectric layer are located directly below the second source field plate.

18. The manufacturing method of the HEMT device according to claim 17, wherein a method of forming the barrier layer, the first dielectric layer, the etch stop layer, and the second dielectric layer comprises: forming a barrier material layer on the channel layer; forming the gate electrode on the barrier material layer; forming a first dielectric material layer on the barrier material layer and the gate electrode; forming an etch stop material layer on the first dielectric material layer; forming a second dielectric material layer on the etch stop material layer; patterning the second dielectric material layer to form the second dielectric layer and to expose the etch stop material layer; and patterning the etch stop material layer, the first dielectric material layer, and the barrier material layer to form the etch stop layer, the first dielectric layer, and the barrier layer and to expose the channel layer.

19. The manufacturing method of the HEMT device according to claim 14, wherein a method of forming the drain electrode, the first source field plate, and the second source field plate comprises: forming a conductive material layer on the channel layer and the dielectric structure; and patterning the conductive material layer to form the drain electrode, the first source field plate, and the second source field plate.

20. The manufacturing method of the HEMT device according to claim 14, further comprising: forming a P-type gallium nitride layer between the gate electrode and the barrier layer.
